Bottu Teja Sai

University of North Carolina Greensboro | TSBOTTU@uncg.edu | +1 (984) 376-1118 | LinkedIn ↗ | GitHub ↗ | Portfolio ↗

PROFESSIONAL SUMMARY

Information Technology graduate student with hands-on experience in AI security, cryptography, and full-stack development. Published researcher in adversarial LLM testing and post-quantum cryptography, with demonstrated ability to design and ship security tools from concept to deployment. Combines a strong foundation in networking and systems with practical experience building microservices, security frameworks, and AI-driven agents.

EDUCATION

University of North Carolina Greensboro 2026 -- Present
Master of Science in Information Technology and Management Greensboro, NC, USA

Velagapudi Ramakrishna Siddhartha Engineering College 2022 -- 2026
Bachelor of Technology in Information Technology Vijayawada, IN
CGPA: 7.11 / 10

RESEARCH & PUBLICATIONS

Adversarial Security Assessment of Large Language Models Through Automated Vulnerability Testing 2026
KalingaConf --- Kalinga Conference on Communication & Computing

Autonomous LLM-Driven Security Agent 2025
ICDECT --- International Conference on Data Engineering and Communication Technology

Dynamic Pattern-Based Text Steganography System 2026
ICCCT --- International Conference on Computing and Communication Technologies

EXPERIENCE

The Offensive Labs Aug 2025 -- Feb 2026
Cybersecurity Developer Remote

- Developed a UI/UX dashboard from scratch, translating design concepts into fully functional interfaces.
- Architected backend server and database systems with a focus on scalability and maintainability.
- Engineered modular microservices for discrete AI agents, improving backend task-handling efficiency.

PROJECTS

AI-GUARDIAN: Secure LLM Defense Framework 2026
Academic and Research Project
Python • LLM Security • Adversarial Testing

- Designed and built a unified security framework to protect Large Language Models from Adversarial Attacks on AI Systems.
- Developed an adversarial attack module with multi-layered detection and defense mechanisms for real-time threat mitigation.
- Reduced attack success rates and enhanced model robustness for secure deployment in real-world generative AI systems.

Post-Quantum Cryptography (PQC) Migration Tool 2025 -- 2026
Internship Project
Python • Cryptography • Security

- Built a scanning system to identify cryptographic algorithms in use across organizational infrastructure.
- Developed a role-based dashboard (Admin, Manager, Developer) to visualize findings and deliver actionable migration insights.
- Classified algorithms as quantum-safe or vulnerable and enabled automated migration to post-quantum secure alternatives.

Autonomous LLM-Driven Security Agent 2025
Research Project
Python • LLM Security • Vulnerability Testing

- Built an AI-powered agent to automate web application vulnerability testing with minimal human involvement.
- Integrated multiple security tools with a local LLM to detect and report common web vulnerabilities.
- Improved testing efficiency by eliminating redundant scans and generating structured, actionable security reports.

SKILLS

Programming Languages: Python
Generative AI: LLM Integration • Adversarial Evaluation • RAG • Fine-tuning (Basics)
Tools & Technologies: Docker • Git • GitHub
Operating Systems: Linux • Windows
Networking Tools: Nmap • Wireshark
Core Competencies: Networking • Cryptography

CERTIFICATIONS

The Bits and Bytes of Computer Networking – Google
CCNA: Intro to Networks – Cisco
Getting Started with Git and GitHub – IBM
Kali Linux Basics – Simplilearn
Linux Commands – Udemy
Joy of Computing – NPTEL
Social networking – NPTEL

ACTIVITIES & LEADERSHIP

National Service Scheme (NSS) – Assistant Director
Led 50+ volunteers, coordinated campus initiatives, and organized community outreach programs.

Association for Computing Machinery (ACM) – Active Member
Contributed to technical workshops and security events.